

Heart Attack Predictions in the United States

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Project Description

Cardiovascular diseases (CVDs) represent a critical public health issue, accounting for the leading cause of mortality globally and exerting a disproportionate burden on the United States healthcare system. According to the Centers for Disease Control and Prevention (CDC), heart disease was responsible for one in every five deaths in the U.S. in 2022, highlighting the need for early detection and risk stratification strategies that can inform preventive interventions [1]. Despite the availability of established risk models, the limitations of these frameworks in accommodating non-linear, high-dimensional data have necessitated the exploration of more robust and adaptive methodologies.

This project employs Apache Spark, a powerful distributed computing framework, in conjunction with machine learning (ML) algorithms to predict heart attack risk using a real-world dataset titled "Heart Attack Prediction in the United States" from Kaggle [2]. The public dataset incorporates structured data on demographic, physiological, and behavioral risk factors, providing an opportunity to model complex interactions that traditional statistical models often overlook.

The rationale for utilizing Spark lies in its capacity to handle large-scale data efficiently, enabling distributed data processing, real-time computation, and integration with MLlib for scalable machine learning. This combination provides a viable approach for transforming static risk scores into dynamic, personalized models that can capture nuanced patterns across diverse populations. This project aims to explore the effectiveness of scalable machine learning models, implemented using Apache Spark, in predicting heart attack risk from a large, real-world U.S. health dataset. By leveraging big data technologies and advanced classification algorithms, the goal is to assess whether such approaches can support accurate and data-driven cardiovascular risk assessment.

Literature Review

1. Evolution of Risk Prediction Models

Traditional cardiovascular risk prediction has largely relied on statistical models derived from longitudinal cohort studies. The Framingham Risk Score (FRS), based on the eponymous heart study, has served as a cornerstone for estimating 10-year coronary heart disease risk [3]. Similarly, the American College of Cardiology/American Heart Association (ACC/AHA) introduced the ASCVD Risk Estimator, which incorporates variables such as age, gender, lipid profiles, and smoking status to assess atherosclerotic cardiovascular disease risk [4]. While these models offer interpretability and ease of implementation, they are constrained by assumptions of linearity and limited capacity to incorporate complex feature interactions or emerging risk markers [5].

2. Superiority of Machine Learning over Conventional Models

Recent advancements have highlighted the potential of machine learning (ML) to outperform traditional models in both discrimination and calibration of cardiovascular risk prediction. Weng et al. (2017) demonstrated that random forests, support vector machines (SVMs), and gradient boosting methods outperformed logistic regression in predicting cardiovascular events using UK general practice data, as measured by net reclassification improvement (NRI) and the area under the curve (AUC) [6]. In a recent study comparing machine learning and conventional statistical models for predicting 30-day re-admission following acute heart failure hospitalization, Abdul-Samad et al. [7] demonstrated that machine learning algorithms, particularly gradient boosting models, achieved superior predictive performance compared to traditional logistic regression approaches.

PremSagar et al. (2022) further validated these findings in a small South African cardiac cohort, demonstrating that machine learning models, such as XGBoost and random forests, outperformed traditional statistical classifiers in terms of precision and recall. While geographically distinct, the study's methodological rigor and consistency with prior evidence underscore the potential generalizability of ML-based approaches across diverse clinical populations [8]. Importantly, these improvements are not merely statistical; they translate to meaningful clinical decision support by reducing both false negatives and false positives.

3. Deep Learning and High-Dimensional Feature Integration

Beyond classical machine learning models, deep learning has emerged as a transformative approach, particularly in processing high-dimensional data, such as imaging, electrocardiograms (ECGs), and electronic health records (EHRs). Rajkomar et al. [9] applied deep neural networks to large-scale electronic health records to predict multiple clinical outcomes, including heart failure and myocardial infarction, achieving predictive performance comparable to that of practicing clinicians across several metrics. In a review of recent advancements, Huang et al. described the integration of wearable sensor data with recurrent neural networks for real-time cardiovascular risk prediction, highlighting a paradigm shift toward continuous and personalized risk assessment [10].

4. Explainability and Clinical Integration

Despite their strong predictive capabilities, machine learning and deep learning models are often criticized for being "black boxes" due to their limited interpretability. In response to this challenge, explainable AI (XAI) techniques have emerged to bridge the gap between accuracy and transparency. One such method is SHapley Additive exPlanations (SHAP), introduced by Lundberg and Lee [11], which unifies several existing feature attribution methods under a single theoretical framework grounded in cooperative game theory. SHAP assigns consistent and locally accurate importance values to each feature, enabling robust interpretation of individual predictions. These properties make SHAP particularly suitable for sensitive domains such as cardiovascular risk prediction, where transparency is crucial for clinical adoption and decision-making.

5. Application in Public Health Datasets and U.S. Context

Lakshmanarao et al. [12] developed a machine learning-based method for heart disease prediction, using ensemble learning and feature selection techniques. They addressed data imbalance using sampling methods and identified predictive features. The models were trained and evaluated on Kaggle and UCI datasets, and among various classifiers tested, ensemble methods showed superior accuracy. The study validated the model's effectiveness across different datasets and highlighted the applicability of open-source tools in clinical predictive modeling. These findings support the use of robust ML pipelines for accurate cardiovascular risk prediction and align with the objectives of the current project.

Data Description

The dataset used in this project was obtained from Kaggle: (<https://www.kaggle.com/datasets/ankushpanday2/heart-attack-prediction-in-united-states>). It is publicly accessible and distributed under an open data license, allowing its use for academic and research purposes. The dataset is provided in CSV format and consists of 372,974 individual health records, each comprising 32 features. These features represent a broad range of variables, encompassing demographic,

clinical, behavioral, and medical history domains. Demographic variables such as age, gender, ethnicity, income, education level, employment status, marital status, and place of residence offer contextual information about the individuals. Clinical features, including cholesterol level, systolic blood pressure, resting and maximum heart rate, body mass index (BMI), ST depression, ECG results, and chest pain type, capture physiological markers relevant to cardiovascular health. Medical history indicators like diabetes, hypertension, smoking status, family history of heart disease, previous heart attack, and stroke history provide insight into predisposing conditions. Behavioral and lifestyle-related variables, such as physical activity level, diet, alcohol consumption, and stress level, are included to assess modifiable risk factors. Additional diagnostic features like medication use, exercise-induced angina, slope of the ST segment, number of major vessels identified by fluoroscopy, and thalassemia status further enrich the dataset. This comprehensive collection of features forms the foundation for developing machine learning models aimed at binary classification of heart attack risk.

Overall, this dataset provides a rich foundation for developing machine learning models aimed at predicting heart disease, supporting the project’s goal of exploring data-driven approaches in cardiovascular risk assessment. There is no temporal element in the dataset; all records are cross-sectional, representing a snapshot in time rather than longitudinal tracking. This makes the dataset appropriate for classification tasks rather than time series or trend analyses.

Table 1 presents a detailed overview of the main continuous features used in the final analysis for the prediction of heart attack. The table includes features like age, cholesterol, blood pressure, heart rate, income, max heart rate, and number of blood vessels. The target variable of the dataset indicates whether the individual has heart disease, which is binary in nature (1 = Yes, 0 = No). In the table, mean, median, min and max values are reported for the continuous variables. The categorical variables were not included in the table because of the uniform distribution between the groups.

Table 1. Data Description of Continuous Variables Used in the Final Analysis

Variables and Labels	Type	N = 372974 Mean, SD Median (IQR) Min-Max
Age (in years)	int64	56.97, 15.87 57 (43-71) 30-84
Cholesterol (in mg/dL)	int64	199.46, 57.73 199 (149-249) 100-299
Blood Pressure (in mm Hg)	int64	134.50, 25.98 134 (112-157) 90-179
Heart Rate (beats per minute)	int64	89.49, 17.34 89 (74-105) 6-119
Income	int64	110033.40, 51961.72 110111 (64957- 155012) 20000-199999

Max Heart Rate	int64	149.46, 28.85 149 (124-174) 100-199
BMI	float64	29.01, 6.35 29 (23.5-34.5) 18-40

The descriptive statistics from key continuous variables show the mean age of the participants to be 57 years. Cholesterol values range between 100 and 299 mg/dL, with a mean of around 199 mg/dL. Systolic blood pressure values vary from 90 to 179 mm Hg, averaging 134 mm Hg. The average resting heart rate is roughly 89 beats per minute, with values spanning from 60 to 119. The average annual income was 110033.40 dollars.

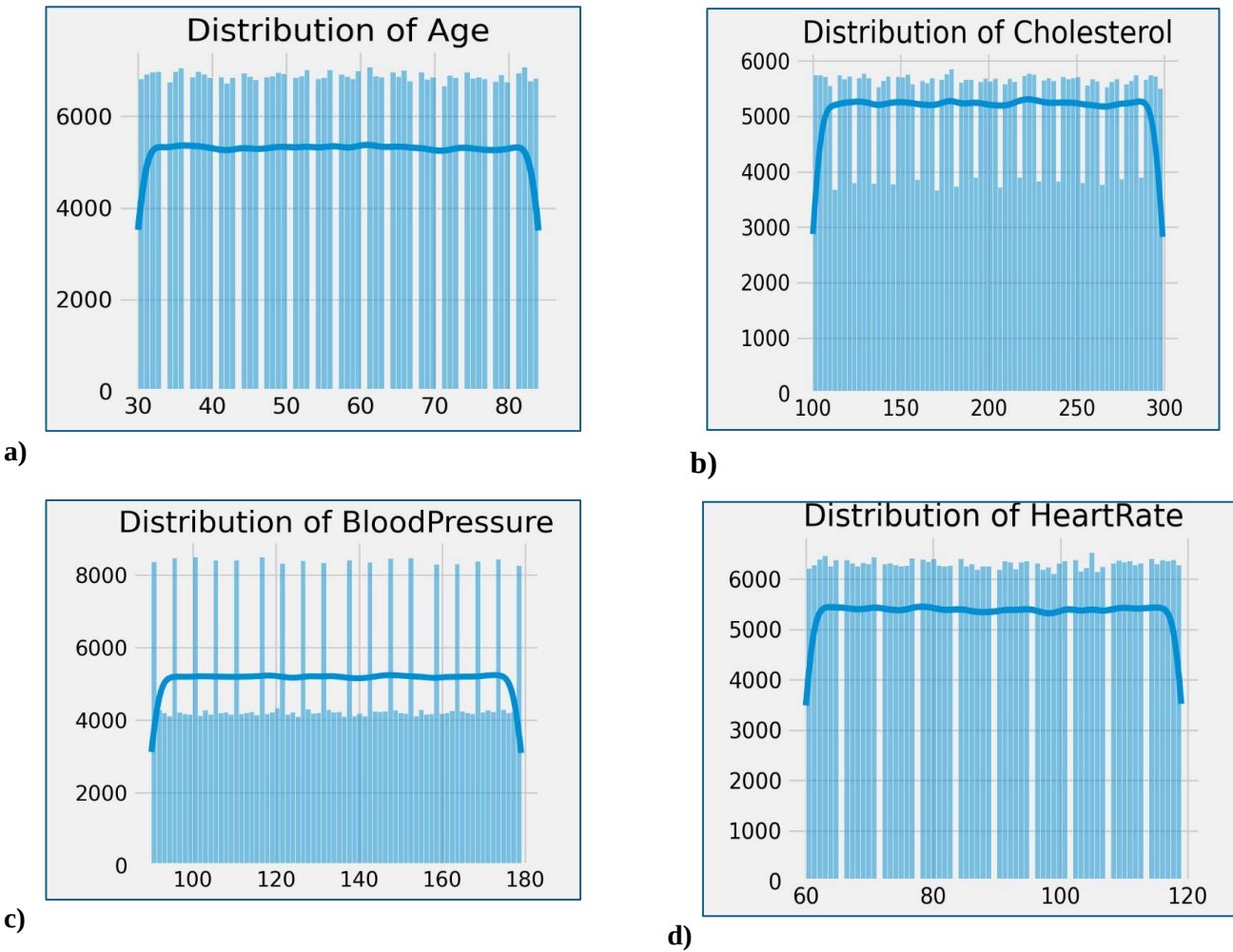


Figure 1: Distributions of some continuous variables

Figure 1 (a, b, c, and d) show the distributions of some continuous variables incorporated into the analysis for this project. The dataset was synthetic and quite perfectly made, which is why we fail to detect any variations among the distributions of the variables.

Methods and Tools

To effectively process and analyze the large-scale dataset comprising over 370,000 records, this project leveraged Apache Spark, a distributed data processing engine optimized for big data workloads. Spark's core advantage lies in its ability to perform in-memory computation across multiple nodes, making it suitable for scalable machine learning through the MLlib library. Given the dataset size and the project's predictive goals, Spark provided both performance efficiency and integration with machine learning pipelines. All model development and data transformation tasks were implemented using PySpark's ML library, specifically modules such as VectorAssembler, StandardScaler, Pipeline, and BinaryClassificationEvaluator.

1. Data Pre-processing

The dataset was first checked for missing values and found to be complete, eliminating the need for imputation. There were no outliers in any of the features. Categorical variables such as Gender, Education Level, and Thalassemia were encoded using Pyspark's StringIndexer to convert string labels into numerical format suitable for model input. Continuous variables including Cholesterol, and MaxHeartRate were standardized using StandardScaler to ensure uniform contribution during model training.

Class balance in the target variable (Outcome) was also assessed. The dataset was found to be nearly perfectly balanced, with approximately 50.05% of records labeled as "No Heart Attack" and 49.95% as "Heart Attack." Given this balance, no additional resampling techniques such as SMOTE (Synthetic Minority Oversampling Technique) were required.

2. Exploratory Data Analysis (EDA)

Exploratory data analysis was performed to understand the underlying structure and distribution of the data. Summary statistics were generated to examine the central tendencies and variability of key variables. To keep the best predictive variables and lower model complexity, feature selection strategies including correlation matrix and random forest were used. A correlation matrix was computed using `Correlation.corr()` from `pyspark.ml.stat`, to assess linear relationships between features and the outcome variable. This observation further justified the use of machine learning models capable of capturing non-linear patterns and feature interactions. Additionally, Python libraries such as *pandas*, *matplotlib*, and *seaborn* were used to visualize distributions, feature correlations, and evaluation metrics like ROC curves.

3. Machine Learning Models

To predict heart attack risk, several supervised classification algorithms were implemented using Apache Spark's MLlib. The model development process began with the selection of both baseline and advanced machine learning models that could handle structured, high-dimensional data. Each model was chosen based on its theoretical strengths, empirical performance in prior cardiovascular studies, and compatibility with distributed processing.

As a baseline, logistic regression was employed due to its interpretability and historical relevance in medical research. This linear model estimates the probability of a binary outcome using a logistic function and serves as a useful benchmark to evaluate the added value of more complex models. While logistic regression offers transparency and ease of explanation, its predictive performance may be limited in the presence of non-linear relationships or interactions among variables. The Random Forest Classifier was used to capture this complexity since it can simulate non-linear interactions and provide interpretability ratings for feature importance.

Model evaluation was based on standard classification metrics, including accuracy, precision, recall, F1-score, and the area under the receiver operating characteristic curve (AUC-ROC). These metrics were computed using PySpark’s BinaryClassificationEvaluator and manual calculations with Spark SQL functions for cross-verification. These metrics were chosen to provide a comprehensive assessment of model performance, particularly in distinguishing between true positives and false positives, an essential consideration in clinical risk prediction.

Data Analysis

1. Workflow Architecture

The entire architecture of the data analysis workflow is summarized in Figure 1. The process begins with reading the raw CSV dataset into Apache Spark. The dataset is subsequently subjected to a series of transformations, including type inference, missing value handling, and categorical encoding using StringIndexer. The resulting features are assembled using VectorAssembler and standardized through StandardScaler to standardize feature contributions and reduce scale-induced bias. These engineered features are then passed to classification models, logistic regression and random forest, for training and evaluation using a reproducible Pipeline architecture.

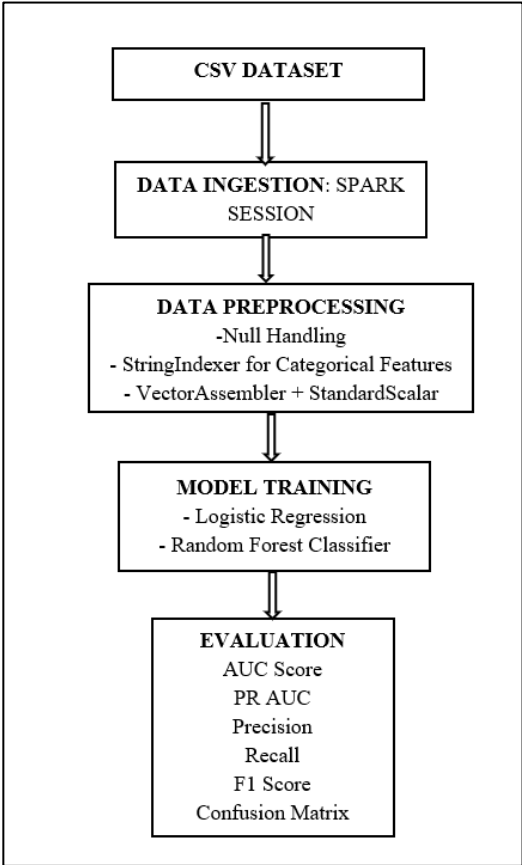


Figure 2: Flow chart of the Workflow Architecture

2. Data Ingestion and Preprocessing

The dataset was imported using Spark’s read.csv function, which automatically parsed the schema. Schema validation was conducted using printSchema() to ensure compatibility with downstream machine learning functions. Missing values in critical numerical and categorical fields were handled using a row-wise deletion strategy to avoid potential

distortions in modeling. All nominal categorical variables (e.g., gender, ethnicity, chest pain type) were encoded into numerical indices using StringIndexer, which facilitated their inclusion in the machine learning workflow.

3. Feature Engineering

Continuous variables like cholesterol and BMI were standardized using z-score standardization to ensure consistent scaling. Domain expertise and exploratory analysis were used to select relevant features, index categorical variables, and combine them into a single vector column (raw_features) using VectorAssembler. A StandardScaler was used to mitigate the impact of factors with varying magnitudes, particularly those such as age, income, and cholesterol. The altered output (final_features) was used for model training, ensuring that no feature dominated due to size rather than predictive power. The relative contribution of each feature to the model is highlighted by the bar plot, which shows the sorted feature importance in Figure 2. However, correlations were generally weak, suggesting that more complex relationships might exist.

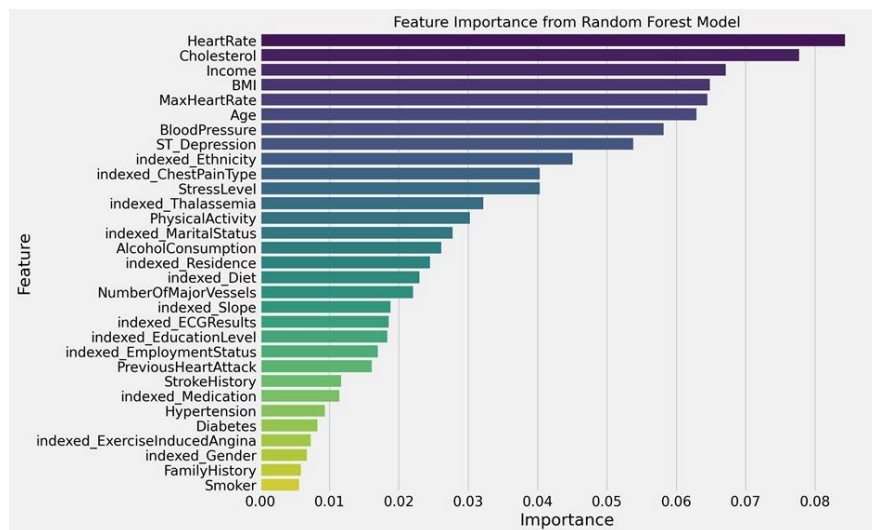


Figure 3: Bar plot of Sorted Feature Importance

4. Model Development and Evaluation

Two classification models were implemented to evaluate the predictability of the outcome variable (indexed_Outcome): logistic regression and random forest classifier. Logistic regression was chosen for its interpretability and baseline utility in binary classification tasks. Random forest was selected to account for potential non-linear interactions between features. Both models were trained on 80% of the data and tested on the remaining 20%, with the BinaryClassificationEvaluator employed to compute the area under the receiver operating characteristic curve (AUC).

Despite rigorous implementation, both models yielded AUC scores near 0.5, suggesting performance indistinguishable from random guessing. According to the correlation matrix in Table 2, this result most likely represents the weak or non-significant correlations between the features and the outcome variable. Any non-linear trends in the data might have been difficult for logistic regression to identify because it assumes linear correlations. However, random forest did not perform as well, perhaps because it overfitted on irrelevant or noisy features. Confusion matrices and predicted

probabilities were examined to further corroborate these findings, which both confirmed the subpar performance and showed that there was no significant signal in the feature space.

Table 2: Confusion matrix between indexed outcome and predicted outcome

Outcome	Predicted Outcome	Count
1.0	1.0	17442
0.0	1.0	17450
1.0	0.0	20460
0.0	0.0	20492

5. Exploratory Analysis and Justification

A correlation heatmap was computed between all input features and the target variable, as shown in Figure 2. Surprisingly, the analysis revealed that none of the features exhibited a meaningful correlation with the outcome, with the highest observed correlation being approximately 0.002. These findings strongly suggested that the dataset lacked intrinsic predictive power for the given label, despite the methodologically sound analysis pipeline.

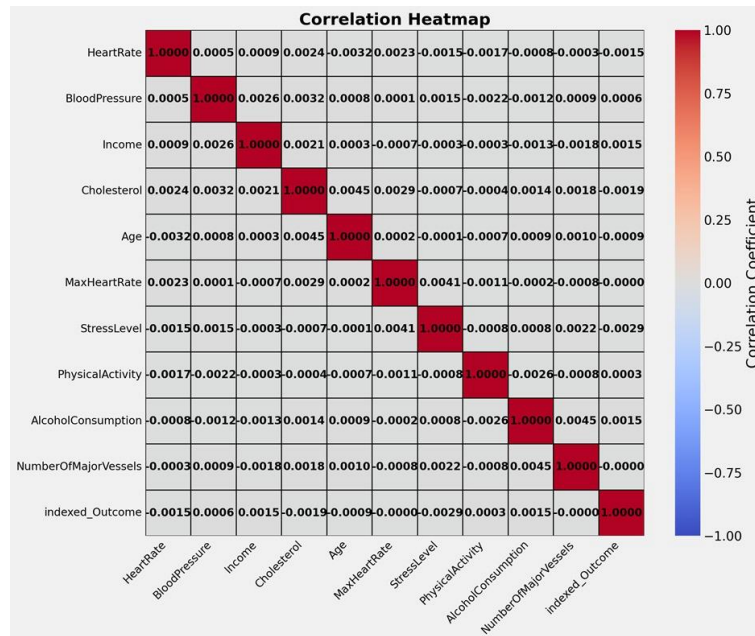


Figure 4: Correlation Heatmap Between Features and Label Variable

Results

The primary objective of this analysis was to evaluate whether a machine learning pipeline could successfully predict heart disease outcomes using behavioral, demographic, and clinical variables. Two classification models, logistic regression and random forest, were trained and evaluated using standardized preprocessing and feature engineering techniques. Figure 3 shows that neither model outperformed random guessing, with area under the receiver operating characteristic curve (AUC) scores at 0.5. The models' incapacity to successfully differentiate between positive and negative examples was further supported by their 0.50 recall, 0.50 accuracy, and 0.50 F1-score.

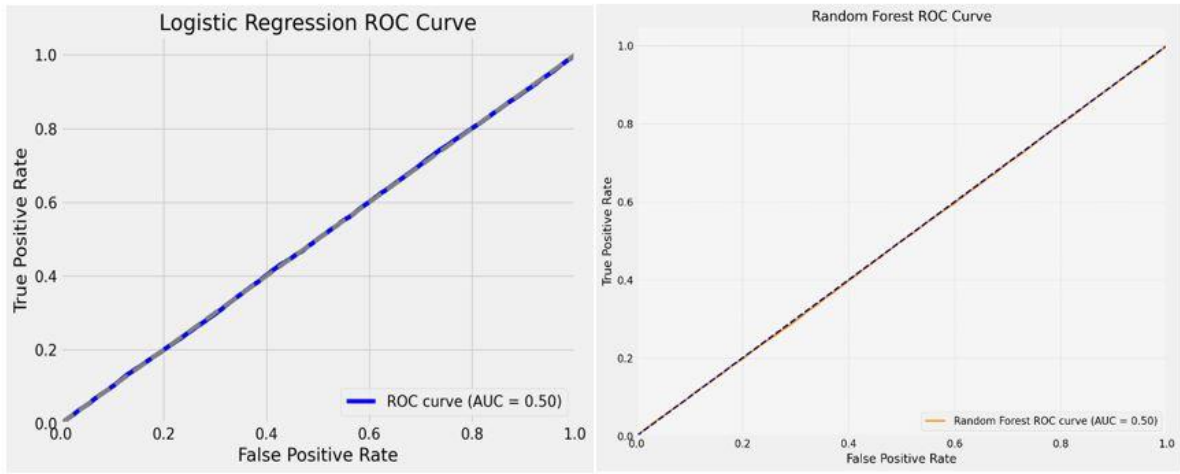


Figure 5: ROC Curve for Logistic Regression vs Random Forest

Further investigation into the data revealed that the lack of predictive power was not due to model underfitting or implementation error but rather a consequence of the intrinsic characteristics of the dataset. Correlation analysis revealed that none of the features exhibited a statistically significant relationship with the target variable (indexed_Outcome).

When compared to prior studies and publicly available datasets (e.g., the UCI Heart Disease dataset), which typically show moderate to strong associations between clinical indicators and heart disease outcomes [13], the present dataset appears to lack a similar predictive structure. This discrepancy highlights a fundamental limitation of the data source itself, rather than the analytical methods employed. While related work has demonstrated success with similar methodologies, those successes were grounded in datasets with clearer signal-to-noise ratios and stronger outcome-feature correlations [5], [14].

Conclusion

This project implemented a complete machine learning workflow to predict heart disease using PySpark, including data ingestion, preprocessing, feature engineering, model training, and evaluation. While the computational infrastructure and methodological design were successful and reproducible, the models consistently failed to achieve meaningful classification performance. The key finding of this study is that the lack of predictive success stemmed not from flaws in modeling but from the absence of signal within the dataset itself.

Several challenges emerged during the analysis. The first was identifying and correcting potential preprocessing pitfalls and encoding categorical variables for Spark compatibility. These were successfully addressed. However, the most significant limitation was the weak relationship between input features and the outcome variable, which severely constrained model learning. This limitation was uncovered through correlation analysis and visual exploration, highlighting the importance of exploratory data analysis as a diagnostic step in predictive modeling.

A key takeaway from this project is that even a well-structured and technically sound machine learning pipeline cannot compensate for a lack of informative features. The experience underscores the crucial role of data quality and feature relevance in achieving predictive success. For future analysis, applying the same pipeline to a more informative dataset, such as the UCI Heart Disease dataset, could better demonstrate its utility. Additionally, incorporating domain expertise

in feature selection or exploring feature interactions might help uncover latent patterns that were not detectable in this dataset.

The study ultimately succeeded in creating a scalable, reproducible, and scientifically sound machine learning pipeline, demonstrating crucial data assessment abilities necessary for practical analysis, despite the models' inability to make predictions.

Contribution to the Report

All four members collaborated on this project, contributing equally and actively to each stage. The team contributed to the project's development, discussion, and overall direction. Regular meetings were held to identify problems, develop solutions, and ensure steady progress. After evaluating several publicly available options, the dataset obtained via Kaggle was selected by mutual consent. Additionally, each member contributed to the report's writing, the model's selection, and the interpretation of the results. The distribution of certain technical and organizational tasks is stated as follows:

1. Data Understanding and Preprocessing

Subhechchha concentrated on comprehending the dataset's structure and preparing it for analysis. Her responsibilities included handling missing values, encoding categorical variables, and ensuring the data were tidy and formatted for modeling. She also made a substantial contribution to the drafting of the report.

2. Exploratory Data Analysis (EDA) and Feature Selection

Zahid oversaw the exploratory data analysis, finding important patterns and associations between heart disease-related characteristics. He also interpreted statistical results and visual insights to aid in the feature selection process.

3. Machine Learning Model Development

Theo took the lead in implementing the machine learning models, even though everyone participated in the discussion about model selection. He was instrumental in coding, adjusting hyperparameters, and evaluating the effectiveness of classifiers, including Random Forest and Logistic Regression.

4. Documentation and Finalization

Filza ensured that the report's methodology, findings, and overall organization were thoroughly documented. She contributed substantially to writing the final report, arranging the findings, and preserving sectional uniformity.

Overall, the project resulted from regular collaboration, delegated tasks, and candid communication among all participants.

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